

ABC SENIOR SECONDARY SCHOOL
CBSE CLASS XII — BOARD EXAMINATION (THEORY)

Subject: Science / Mathematics

Time Allowed: 3 Hours | Maximum Marks: 80 | Date: _____

Name: _____ Roll No.: _____

Class/Section: _____

GENERAL INSTRUCTIONS

- 1. All questions are compulsory.*
- 2. Read each question carefully before answering.*
- 3. Figures to the right indicate full marks.*
- 4. Use of calculator is not permitted unless stated.*

SECTION A — MULTIPLE CHOICE QUESTIONS

Each question carries 1 mark. Select the correct option.

Q1. (1 marks)

What happens when a fault occurs in the electrical system?

- (a) The MCB will trip and interrupt the current flow
- (b) A fuse will melt, breaking the circuit
- (c) The system will automatically repair itself
- (d) The voltage will increase to stabilize the system

Study the figure below.

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Q2. (1 marks)

Which of the following statements correctly describe the behavior of a fuse and a MCB in an electrical circuit?

- (a) Only the fuse will melt, breaking the circuit
- (b) Only the MCB will trip and interrupt the current flow
- (c) Both the fuse and MCB will behave the same way
- (d) The correct options cannot be determined without specific figures

Study the figure below.

CT UNIVERSITY
END SEMESTER EXAMINATION

Subject: GA
Time Allowed: 3 Hours | Maximum Marks: 70 | 25 May 2026

Name: _____ Roll No.: _____
Class/Section: _____

GENERAL INSTRUCTIONS
1. Answer all questions.
2. Part A is compulsory.
3. In Part B, answer any 4 out of 6.
4. In Part C, answer any 2 out of 3.

PART A — OBJECTIVE (COMPULSORY)

Answer all 10 questions. 1 mark each.

Q1. (1 marks)

What happens when a fault occurs in the electrical system?

- (a) The MCB will trip and interrupt the current flow
- (b) A fuse will melt, breaking the circuit

Study the figure below.

Q2. (1 marks)

Q3. (1 marks)

Which of the following statements accurately describe the railway signal model depicted in the figure?

- (a) All four statements are accurate descriptions
- (b) Only two statements accurately describe aspects related to fuses and MCB
- (c) None of the statements accurately describe the model
- (d) The correct option cannot be determined without referring to specific figures

Study the figure below.

Which statement about a concave lens' ability to form images is true?
Q3. (1 marks)
If an object is placed at twice the focal length from a concave mirror, what can be said about the image formed?
(a) The image will also be real and inverted
(b) No image will form
Q4. (1 marks)
A terminal is the point at which two conductors are connected to an electrical circuit. Which of these options best describes a device that connects multiple cells in series or parallel?
(a) Battery
(b) Resistor
(c) Capacitor
(d) Diode
Study the figure below.

Q5. (1 marks)
When current flows through a room heater, it encounters resistance. This results in the generation of heat to warm up the air inside the space around the appliance. Which device is specifically designed for this purpose?
(a) Resistor
(b) Transformer
(c) Thermostat
(d) Fuse

Q4. (1 marks)

Which of the following best describes the function of a capacitor in an electrical circuit?

- (a) Stores electrical energy in an electric field
- (b) Stores electrical energy in a magnetic field
- (c) Serves as a switch to control current flow
- (d) Is used to increase the voltage

Study the figure below.

Q5. (1 marks)

What is the primary purpose of a resistor in an electronic circuit?

- (a) To amplify the signal
- (b) To convert AC to DC
- (c) To regulate the current flow
- (d) To store electrical energy

Study the figure below.

Q6. (1 marks)

Which of the following statements is true regarding the relationship between voltage, current, and resistance in an electrical circuit?

- (a) Voltage is the rate at which electrical energy is used
- (b) Current is the resistance to the flow of electrons
- (c) Resistance is the product of voltage and current
- (d) Ohm's Law states that $V = I \times R$

Study the figure below.

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Q7. (1 marks)

If an object is placed at twice the focal length from a concave mirror, what can be said about the image formed?

- (a) The image will also be real and inverted
- (b) No image will form
- (c) The image will be virtual and upright
- (d) The image will be the same size as the object

Q8. (1 marks)

Which of these options best describes a device that connects multiple cells in series or parallel?

- (a) Battery
- (b) Resistor
- (c) Capacitor
- (d) Diode

SECTION B — VERY SHORT ANSWER

Answer in 40-60 words. Each question carries 2 marks.

Q9. (2 marks)

Based on the principles of motion and velocity, describe the difference between uniform and non-uniform motion. ?

Q10. (2 marks)

Arrange the following activities in order of increasing breathing rates: sleeping, walking, running, and sprinting. ?

SECTION C — SHORT ANSWER

Answer in 60-80 words. Each question carries 3 marks.

Q11. (3 marks)

Explain how resistance in a circuit leads to the generation of heat, using an example from everyday life.

Study the figure below.

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Q12. (3 marks)

Identify and explain the function of a device that is specifically designed to generate heat from electrical current, and give an example of such a device.

SECTION D — CASE BASED / SOURCE BASED

Read the passage and answer the sub-questions. Each question carries 4 marks.

Q13. (4 marks)

Refer to the diagram below for the following questions. The diagram shows a simple electrical circuit with a battery, a fuse, and a heater connected in series.

A simple electrical circuit with a battery, a fuse, and a heater connected in series.

- (a) What is the primary function of the fuse in the circuit?
- (b) If the heater is turned on and the current flows through it, what phenomenon is primarily responsible for the heating effect?

SECTION E — LONG ANSWER

Answer in detail with diagrams where necessary.

Q14. (5 marks)

Given the distance-time graph for the motion of two vehicles A and B as shown in the diagram, vehicle A completes 20 full oscillations in 32 seconds, and the railway track between City X and City Y is 240 kilometers long, which the train travels in 4 hours without any stops. Calculate the time period 'T' of a single oscillation for the pendulum and the constant speed 'v' of the train. Additionally, identify which vehicle (A or B) shows a truck moving with a non-constant speed based on the graph. (5 marks)

Figure

SECTION F — EXTENDED RESPONSE

Answer with proper derivation / explanation. (8 marks)

Q15. (8 marks)

Refer to the distance-time graph provided in Fig_0001, which shows the motion of two vehicles A and B. Vehicle A starts from rest and accelerates uniformly, while vehicle B maintains a constant speed throughout its journey. Calculate the acceleration of vehicle A and its speed at the moment it overtakes vehicle B, and explain your reasoning. ?

Figure